

Practical-09

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GNU nano 8.7.1                                file_copy.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[])
{
    int source_fd, dest_fd;
    ssize_t bytes_read, bytes_written;
    char buffer[BUFFER_SIZE];

    if (argc != 3)
    {
        printf("Usage: %s <source_file> <destination_file>\n", argv[0]);
        return 1;
    }

    source_fd = open(argv[1], O_RDONLY);

    if (source_fd < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    dest_fd = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (dest_fd < 0)
    {
        perror("Error opening destination file");
        close(source_fd);
        return 1;
    }

    while ((bytes_read = read(source_fd, buffer, BUFFER_SIZE)) > 0)
    {
        bytes_written = write(dest_fd, buffer, bytes_read);

        if (bytes_written != bytes_read)
        {
            perror("Error writing to destination file");
            close(source_fd);
            close(dest_fd);
            return 1;
        }
    }

    if (bytes_read < 0)
    {
        perror("Error reading source file");
    }

    lseek(source_fd, 0, SEEK_SET);

    close(source_fd);
    close(dest_fd);

    printf("File copied successfully using low-level system calls.\n");

    return 0;
}
```

Output:

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical$ cd Practical-09
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ nano file_copy.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ gcc file_copy.c -o file_copy
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ echo "This is Practical 9 file copy demonstration." > source.txt
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat source.txt
This is Practical 9 file copy demonstration.
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ gcc file_copy.c -o file_copy
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ ls
file_copy  file_copy.c  source.txt
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat copied.txt
cat: copied.txt: No such file or directory
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ ls -l source.txt copied.txt
ls: cannot access 'copied.txt': No such file or directory
-rw-r--r-- 1 uday uday 45 Oct  1 09:24 source.txt
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat source.txt
This is Practical 9 file copy demonstration.
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ |
```

Part B :

```
GNU nano 7.2 io_redirect.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int fd;

    fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0)
    {
        perror("Error opening file");
        return 1;
    }

    if (dup2(fd, STDOUT_FILENO) < 0)
    {
        perror("dup2 failed");
        close(fd);
        return 1;
    }

    close(fd);

    printf("This output is redirected from standard output to output.txt\n");
    printf("dup2() successfully redirected stdout.\n");

    return 0;
}
```

Output:

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ nano io_redirect.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ gcc io_redirect.c -o io_redirect
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ ls
file_copy file_copy.c io_redirect io_redirect.c source.txt
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ ./io_redirect
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat output.txt
This output is redirected from standard output to output.txt
dup2() successfully redirected stdout.
```

Part c :

Output:

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ nano file_copy.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ gcc file_copy.c -o file_copy
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ echo "This is Practical 9 file copy demonstration." > source.txt
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat source.txt
This is Practical 9 file copy demonstration.
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ gcc file_copy.c -o file_copy
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ ls
file_copy file_copy.c source.txt
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat copied.txt
cat: copied.txt: No such file or directory
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ ls -l source.txt copied.txt
ls: cannot access 'copied.txt': No such file or directory
-rw-r--r-- 1 uday uday 45 Oct 1 09:24 source.txt
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat source.txt
This is Practical 9 file copy demonstration.
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ nano io_redirect.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ gcc io_redirect.c -o io_redirect
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ ls
file_copy file_copy.c io_redirect io_redirect.c source.txt
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ ./io_redirect
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat output.txt
This output is redirected from standard output to output.txt
dup2() successfully redirected stdout.
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ echo "Hello from shell output redirection" > shell_output.txt
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat shell_output.txt
Hello from shell output redirection
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ echo "This is input redirection using the shell." > input.txt
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat < input.txt
This is input redirection using the shell.
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat < input.txt
This is input redirection using the shell.
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ cat final_output.txt
This is input redirection using the shell.
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ ls -l
total 60
-rwxr-xr-x 1 uday uday 16304 Oct 1 09:24 file_copy
-rw-r--r-- 1 uday uday 1286 Oct 1 09:20 file_copy.c
-rw-r--r-- 1 uday uday 43 Oct 1 09:31 final_output.txt
-rw-r--r-- 1 uday uday 43 Oct 1 09:30 input.txt
-rwxr-xr-x 1 uday uday 16128 Oct 1 09:28 io_redirect
-rw-r--r-- 1 uday uday 541 Oct 1 09:28 io_redirect.c
-rw-r--r-- 1 uday uday 100 Oct 1 09:28 output.txt
-rw-r--r-- 1 uday uday 36 Oct 1 09:30 shell_output.txt
-rw-r--r-- 1 uday uday 45 Oct 1 09:24 source.txt
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Practical/Practical-09$ |
```